

SECTION 1: STATUS OF FARM MECHANIZATION IN INDIA

1.1 Definition and Scope

Farm mechanization is the application of engineering and technology inputs (mechanical, electrical, and renewable power sources with matching implements) to perform agricultural operations in a timely and efficient manner, reducing drudgery and cost while raising productivity.

Benefits: timeliness of operations (saves 15–20% of inputs like seed and fertilizer), reduction of drudgery, precision in metering and placement, higher cropping intensity, reduction in cost of operation per unit output, and enabling conservation agriculture.

1.2 Farm Power Availability — Trends (memorize these figures)

Year	Farm power availability (kW/ha)
1951–52	~0.25
1975–76	0.48
2013–14	1.84
2018–19	2.49
2019–20	2.54
2021–22	~3.04
Target 2030	4.0

- There is a strong positive correlation between farm power availability (kW/ha) and food-grain productivity (t/ha) across Indian states — Punjab and Haryana have the highest power availability and productivity.
- **Overall level of farm mechanization in India: ~45–47%**, against ~95% in the USA, ~75% in Brazil, and ~57–60% in China.

1.3 Shift in Power Sources (animate → mechanical)

- 1960–61: animate sources (human + draught animal) supplied ~93–97% of farm power.
- 2018–19 onward: **mechanical + electrical sources supply >90%** of total farm power.
- Approximate present shares of farm power: **tractors ~47–50%, electric motors ~25–27%, diesel engines ~8–9%, draught animals ~5%, agricultural workers ~5%, power tillers ~1%.**
- Tractor is the single largest source of farm power in India.

1.4 Operation-wise Mechanization Levels (approximate)

Operation	Mechanization level
Soil working & seedbed preparation	~40%
Seeding & planting	~29–30%
Plant protection	~34–35%
Irrigation	~37%
Harvesting & threshing (wheat, rice)	60–70%
Harvesting (other crops)	<15–20%

Rice transplanting, cotton picking, sugarcane harvesting, and horticultural operations remain the least mechanized.

1.5 Tractor Industry

- **India is the world's largest tractor manufacturer** and market; domestic sales crossed **~1.0 million units in 2025** (about one-third of global production).
- Major manufacturers: Mahindra & Mahindra (largest, ~40%+ share), TAFE, Sonalika (ITL), Escorts Kubota, John Deere India, New Holland.
- Most-sold power band in India: **31–40 hp and 41–50 hp** categories.
- Tractor density is highest in the north-western states (Punjab, Haryana, western UP).

1.6 Government Programmes and Institutions

- **SMAM (Sub-Mission on Agricultural Mechanization)** — launched **2014–15**;

components include Custom Hiring Centres (CHCs), Hi-Tech Hubs, Farm Machinery Banks, subsidies of 40–50% (up to 80% for CHC/group ownership), training and testing.

- **CRM Scheme** (Promotion of Agricultural Mechanization for In-Situ Management of Crop Residue) — 2018; promotes Happy Seeder, Super SMS, mulcher, zero-till drill, balers to curb residue burning in Punjab, Haryana, UP, NCT Delhi.
- **FARMS mobile app** (Farm Machinery Solutions) — 2019 — Uberisation of custom hiring.
- **Farm Machinery Training & Testing Institutes (FMTTIs)** — 4 institutes:
 - **Budni (Madhya Pradesh)** — Central FMTTI, established **1955**; tractor testing (national tractor test code).
 - **Hisar (Haryana)** — Northern Region FMTTI, 1963.
 - **Garladinne, Anantapur (Andhra Pradesh)** — Southern Region FMTTI, 1983.
 - **Biswanath Chariali (Assam)** — North-Eastern Region FMTTI, 1990.
- **CIAE Bhopal** (Central Institute of Agricultural Engineering, est. 1976, ICAR) — R&D on farm equipment; AICRP on Farm Implements and Machinery (FIM), AICRP on Ergonomics & Safety (ESA), AICRP on Utilization of Animal Energy.
- **Dangerous Machines (Regulation) Act, 1983** — covers power threshers, chaff cutters, sugarcane crushers (operator safety).